

# Data Flow Diagram (DFD)

## AI Learning Assistant

### Project Title

AI Learning Assistant

### Domain

Artificial Intelligence | Education Technology

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## 1. Introduction

A Data Flow Diagram (DFD) is a graphical representation of how data moves through a system. It shows the flow of information between users, processes, data storage, and external systems. The DFD helps in understanding how the application processes user input and generates meaningful output.

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## 2. Purpose of DFD

The Data Flow Diagram is designed to:

- Illustrate the flow of information within the system.
  - Show interactions between the user and the AI Learning Assistant.
  - Explain how user input is processed.
  - Demonstrate how responses and logs are generated and stored.
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## 3. Components of the DFD

### External Entity

- Student/User

## Processes

1. User Input
2. Emotion Detection
3. AI Response Generation (Gemini AI)
4. Log Storage
5. Dashboard Analytics

## Data Store

- Emotion Logs (CSV File)

## Output

- Personalized Study Guidance
  - Emotion Analytics Dashboard
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# 4. Data Flow Description

### Step 1

The student enters a study-related message or query into the application.

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### Step 2

The Emotion Detection module analyzes the text and identifies the student's emotional state (such as Happy, Sad, Stressed, Confused, etc.).

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### Step 3

The detected emotion and user query are sent to the Gemini AI module.

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### Step 4

Gemini AI generates a personalized response based on the detected emotion.

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## Step 5

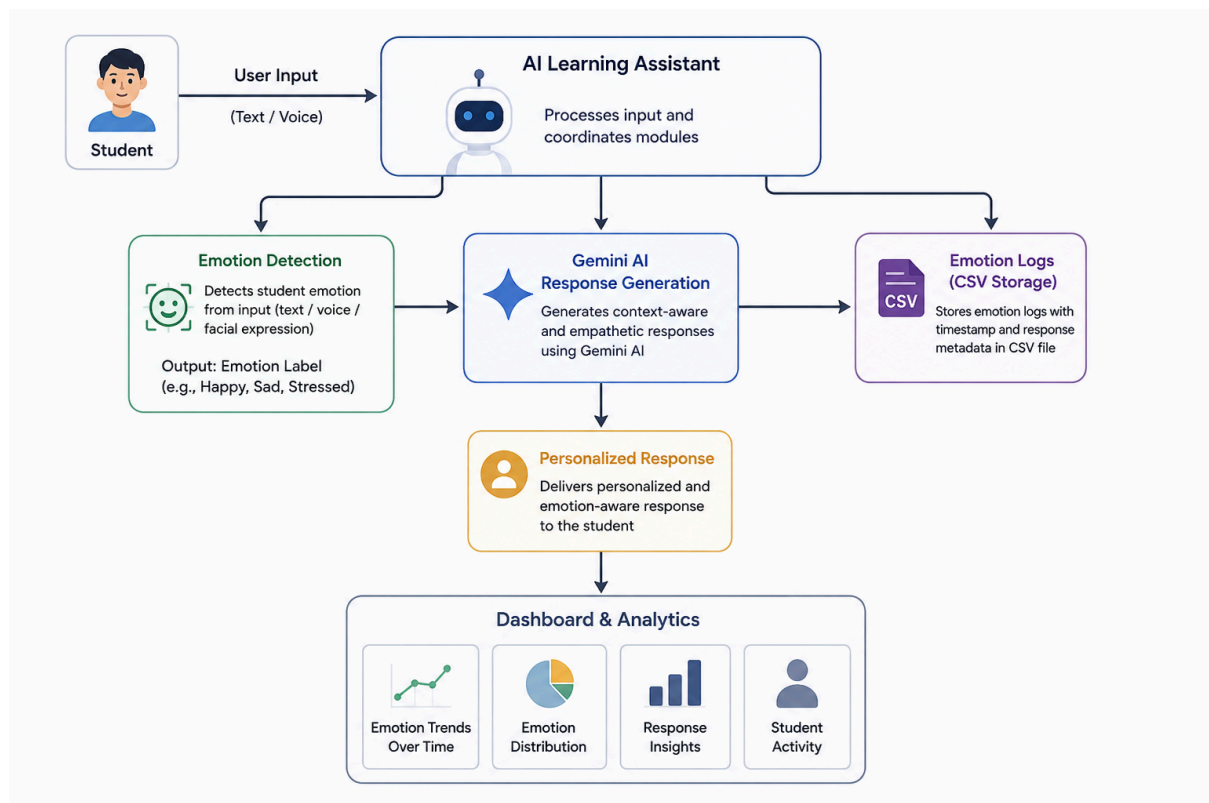
The interaction details (timestamp, user input, detected emotion, confidence score, and AI response) are stored in the Emotion Logs CSV file.

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## Step 6

The Dashboard reads the stored logs and displays analytics such as emotion distribution and interaction history.

# 5. Level-0 DFD



## 6. Explanation

The student interacts with the AI Learning Assistant by entering text. The system analyzes the emotional state using the Emotion Detection module. Based on the detected emotion, Gemini AI generates personalized learning guidance. Every interaction is stored in a CSV file, which is later used by the dashboard to display analytical reports and emotion trends.

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## 7. Advantages of the DFD

- Clearly represents system workflow.
  - Simplifies understanding of data movement.
  - Helps during system design and implementation.
  - Improves documentation quality.
  - Assists in identifying system components and interactions.
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## 8. Conclusion

The Data Flow Diagram provides a clear understanding of how data moves through the AI Learning Assistant. It illustrates the interaction between the student, emotion detection module, Gemini AI, data storage, and dashboard analytics, ensuring a structured and efficient system design.